

Lecture 6

Application I: point groups and molecular vibrations

Objectives. *The machinery of Lectures 3–5 now earns its keep on a concrete physical problem: the small oscillations of a molecule about equilibrium. We set up the dynamical matrix from first principles, let the point group act on displacements, and prove the central theorem of the subject: symmetry block-diagonalises the dynamical matrix, so that vibrational frequencies organise into families labelled by irreducible representations. Their dimensions give symmetry-enforced multiplet sizes; reduced-matrix multiplicity and the reality of the physical displacement space determine the total degeneracy. The programme is carried out in full for water and ammonia, including explicit symmetry-adapted modes from projection operators.*

6.1 Small oscillations and the dynamical matrix

Consider a molecule of N atoms with masses m_a and equilibrium positions $\vec{R}_a^0$, $a = 1, \dots, N$, interacting through a potential V that depends only on the instantaneous positions $\vec{R}_a = \vec{R}_a^0 + \vec{u}_a$. Equilibrium means the gradient vanishes there, so for small displacements Taylor's theorem gives

$$V = V_0 + \frac{1}{2} \sum_{\alpha\alpha, \beta\beta} \left. \frac{\partial^2 V}{\partial u_{a\alpha} \partial u_{b\beta}} \right|_0 u_{a\alpha} u_{b\beta} + O(u^3),$$

with $\alpha, \beta \in \{x, y, z\}$ Cartesian indices: to leading order, every molecule is a system of coupled harmonic oscillators. Newton's equations $m_a \ddot{u}_{a\alpha} = -\partial V / \partial u_{a\alpha}$ carry awkward masses; they disappear in the *mass-weighted coordinates* $q_{a\alpha} = \sqrt{m_a} u_{a\alpha}$, in terms of which

$$\ddot{q} = -D q, \quad D_{a\alpha, b\beta} = \frac{1}{\sqrt{m_a m_b}} \left. \frac{\partial^2 V}{\partial u_{a\alpha} \partial u_{b\beta}} \right|_0. \quad (6.1)$$

The $3N \times 3N$ real symmetric matrix D is the *dynamical matrix*. (The symbol D is entrenched in physics for this object, as $D(g)$, $D^{(i)}(g)$ are for the representation matrices of Lectures 3–5; the two will never meet in one formula, and an argument g or a superscript always signals the latter.)

Being real symmetric, D has an orthonormal eigenbasis, $D \epsilon^{(k)} = \omega_k^2 \epsilon^{(k)}$, and the general solution of (6.1) is a superposition of *normal modes*

$$q(t) = \epsilon^{(k)} \cos(\omega_k t + \varphi_k) :$$

collective motions in which every atom oscillates with one common frequency ω_k and fixed amplitude pattern $\epsilon^{(k)}$. At a stable equilibrium $D \geq 0$, so all $\omega_k^2 \geq 0$; modes with $\omega_k = 0$ will turn out to be not vibrations at all but rigid motions (Section 6.3). The questions this lecture answers by pure symmetry: how do the $3N$ modes organise, which frequencies are forced to coincide, and what do the patterns $\epsilon^{(k)}$ look like — all *before* any spring constant is known.

6.2 The displacement representation

The configuration of small displacements is the vector $u = (\vec{u}_1, \dots, \vec{u}_N)$, an element of the $3N$ -dimensional real *displacement space* $V_{\text{disp}, \mathbb{R}}$. We apply the complex character theory of Lectures 3–5 to its complexification $V_{\text{disp}} = V_{\text{disp}, \mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$. Complex conjugation C on this space fixes the physical real vectors, commutes with the real point-group matrices and with the real dynamical matrix, and will therefore relate conjugate complex irreducibles. The point group G of the equilibrium configuration acts on this space, and the action is an instance — the course's third — of the function-space construction of Lecture 3. A displacement field is a vector-valued function on the set of atoms, $a \mapsto \vec{u}_a$; a symmetry operation $g \in G \subset O(3)$, with 3×3 matrix $R(g)$, moves atom a to the site of atom $g(a)$ (a permutation of identical nuclei) and rotates displacement vectors. The transported field assigns to the *new* site the rotated displacement of the *old*:

$$(\rho(g)u)_a = R(g) \vec{u}_{g^{-1}(a)}, \quad (6.2)$$

the familiar inverse on the argument, with the new feature that the value is rotated too. The homomorphism property follows exactly as in Lecture 3. Since g maps each atom to one of equal mass, the matrices of $\rho(g)$ are the same in mass-weighted as in plain coordinates, and they are orthogonal (a permutation of blocks, each block the orthogonal $R(g)$).

Everything we need from this representation is its character, and the fixed-point lemma of Lecture 5 delivers it instantly, upgraded by the 3×3 trace:

Lemma 6.1 (Atoms left in place). *Let N_g be the number of atoms left in place by the operation g (atoms lying on its axis or plane). Then*

$$\chi_{\text{disp}}(g) = N_g \cdot \chi_{\text{vec}}(g), \quad \text{where} \quad \chi_{\text{vec}}(g) = \text{tr } R(g) = \begin{cases} 1 + 2 \cos \theta, & g = C(\theta), \\ -1 + 2 \cos \theta, & g = S(\theta), \end{cases}$$

with $C(\theta)$ a proper rotation through θ and $S(\theta)$ an improper operation (rotoreflexion).

Proof. In the basis labelled by (atom, direction), the matrix of $\rho(g)$ has 3×3 blocks: by (6.2), block row a , block column b contains $R(g) \delta_{a, g(b)}$. Diagonal blocks occur exactly for atoms with $g(a) = a$, each contributing $\text{tr } R(g)$; moved atoms contribute nothing to the trace. For the 3×3 traces: a proper rotation through θ has, in a basis adapted to its axis, the matrix $\text{diag}(R_{2 \times 2}(\theta), 1)$ with trace $1 + 2 \cos \theta$; an improper operation is, by Lecture 2's factorisation $O(3) = SO(3) \times \{\pm \mathbb{I}\}$, of the form $-C(\pi - \dots)$ — more directly, every improper operation is a *rotoreflexion* $S(\theta)$, rotation through θ composed with reflection in the perpendicular plane, $\text{diag}(R_{2 \times 2}(\theta), -1)$, of trace $2 \cos \theta - 1$. The trace is basis-independent, so the adapted basis costs nothing. ■

The values needed in practice:

g	E	C_2	C_3	C_4	C_6	$\sigma = S(0)$	$i = S(\pi)$
$\chi_{\text{vec}}(g)$	3	-1	0	1	2	1	-3

6.3 Discarding rigid motions

Six of the $3N$ degrees of freedom are not vibrations. A rigid translation, $\vec{u}_a = \vec{t}$ for all a , and an infinitesimal rigid rotation, $\vec{u}_a = \delta \vec{\omega} \times \vec{R}_a^0$, change no interatomic distance, hence

leave V *exactly* invariant: they are zero modes of D , with $\omega = 0$, and they span invariant subspaces of the displacement representation (a symmetry operation maps rigid motions to rigid motions). The genuine vibrations live in the complementary invariant subspace, and at the level of characters the separation is a subtraction:

$$\chi_{\text{vib}} = \chi_{\text{disp}} - \chi_{\text{trans}} - \chi_{\text{rot}}.$$

The two subtracted characters are easy. Translations are labelled by the vector $\vec{t}$, which transforms by $R(g)$ itself: $\chi_{\text{trans}} = \chi_{\text{vec}}$. Rotations are labelled by $\delta\vec{\omega}$, which is an *axial* vector, and axial vectors acquire a determinant:

Lemma 6.2 (Axial vectors). For $R \in \text{O}(3)$, $(R\vec{a}) \times (R\vec{b}) = \det(R) R(\vec{a} \times \vec{b})$. Consequently $\delta\vec{\omega} \mapsto \det(R(g))R(g)\delta\vec{\omega}$, and

$$\chi_{\text{rot}}(g) = \det(R(g)) \chi_{\text{vec}}(g) = \begin{cases} 1 + 2 \cos \theta, & g \text{ proper,} \\ 1 - 2 \cos \theta, & g \text{ improper.} \end{cases}$$

Proof. For proper R the identity is geometry: rotations preserve lengths, angles and orientation, and the cross product is defined by exactly those data, so it commutes with rotations. A general orthogonal matrix is R or $-R$ with R proper (Lecture 2), and $(-\vec{a}) \times (-\vec{b}) = \vec{a} \times \vec{b}$ supplies the factor $\det(-\#) = -1$ relative to the polar transformation law. Writing $\vec{u}_a = \delta\vec{\omega} \times \vec{R}_a^0$ and transforming both the displacement (polar) and the sites, consistency forces $\delta\vec{\omega}$ to transform with the extra det. ■

For a *nonlinear* molecule the rotation labels $\delta\vec{\omega}$ fill three dimensions and

$$\dim V_{\text{vib}} = 3N - 6;$$

a linear molecule has only two independent rotations (spinning about its own axis moves nothing) and $3N - 5$ vibrations — a caveat that matters for CO_2 .

6.4 The central theorem: symmetry block-diagonalises D

That G is a symmetry of the molecule — it permutes identical nuclei and preserves all interatomic geometry — means precisely that the potential is invariant: $V(\rho(g)u) = V(u)$ for all $g \in G$ and all displacements. For the quadratic form $V = \frac{1}{2} q^T D q$ this reads $\rho(g)^T D \rho(g) = D$, and since the matrices $\rho(g)$ are orthogonal,

$$[D, \rho(g)] = 0 \quad \text{for all } g \in G. \quad (6.3)$$

The dynamical matrix is an intertwiner of the displacement representation with itself — and Lectures 4–5 have told us exactly how much freedom such an operator has.

Theorem 6.3 (Symmetry structure of the vibrational spectrum). Decompose the vibrational representation into irreducibles, $V_{\text{vib}} \cong \bigoplus_i m_i V^{(i)}$, and choose in each isotypic component V_i a symmetry-adapted orthonormal basis: vectors $w_\mu^{(a)}$, $a = 1, \dots, m_i$, $\mu = 1, \dots, n_i$, each copy transforming with the same unitary matrices, $\rho(g) w_\nu^{(a)} = \sum_\mu D_{\mu\nu}^{(i)}(g) w_\mu^{(a)}$, and $\langle w_\mu^{(a)}, w_\nu^{(b)} \rangle = \delta_{ab} \delta_{\mu\nu}$. Then:

- (i) D preserves every isotypic component, and has no matrix elements between different components or different partner indices:

$$\left\langle w_\mu^{(a)}, D w_\nu^{(b)} \right\rangle = F_{ab}^{(i)} \delta_{\mu\nu},$$

with an $m_i \times m_i$ Hermitian reduced matrix $F^{(i)}$ that is independent of the partner index μ ;

- (ii) if an eigenvalue ω^2 of $F^{(i)}$ has multiplicity q , then its total complex degeneracy in this isotypic component is qn_i ; the factor $n_i = \dim_{\mathbb{C}} V^{(i)}$ is the symmetry-enforced multiplet size;
- (iii) if every reduced eigenvalue is simple and eigenvalues from unrelated reduced blocks do not coincide, the complex multiplets have size n_i . Repeated reduced eigenvalues contribute the factor q , while coincidences between unrelated blocks are accidental. For a real physical system the conjugate-pair constraints below may enforce additional pairing.

Proof. D commutes with each projector P_i , which is a linear combination of the $\rho(g)$ (Lecture 5); hence $DP_i = P_i D$ and D maps $V_i = \text{im } P_i$ to itself, with $P_j DP_i = DP_j P_i = 0$ for $j \neq i$: no cross-component matrix elements. Within V_i , expand

$$D w_\nu^{(b)} = \sum_{a, \mu} K_{\mu\nu}^{ab} w_\mu^{(a)}.$$

Apply $\rho(g)$ to both sides and use (6.3) together with the adapted transformation law; comparing coefficients of $w_\mu^{(a)}$ gives, for every g and every pair (a, b) ,

$$\sum_{\lambda} K_{\mu\lambda}^{ab} D_{\lambda\nu}^{(i)}(g) = \sum_{\lambda} D_{\mu\lambda}^{(i)}(g) K_{\lambda\nu}^{ab} :$$

each $n_i \times n_i$ matrix K^{ab} commutes with the irreducible representation $D^{(i)}$, hence by Schur's lemma $K^{ab} = F_{ab} \mathbb{1}$, i.e. $K_{\mu\nu}^{ab} = F_{ab} \delta_{\mu\nu}$. This is (i); Hermiticity of $F^{(i)}$ is inherited from D .

For (ii), diagonalise $F^{(i)}$: for each vector c in a q -dimensional ω^2 -eigenspace, the vectors $\sum_b c_b w_\mu^{(b)}$ for $\mu = 1, \dots, n_i$ are n_i linearly independent eigenvectors of D with the common eigenvalue ω^2 ; varying c gives qn_i independent eigenvectors — one for each partner index, all with the same reduced eigenvalue because $F^{(i)}$ did not depend on μ . Conversely every eigenvector of D in V_i arises this way, since the $w_\mu^{(a)}$ exhaust V_i . Part (iii) separates the generic simplicity hypothesis from the two possible kinds of coincidence. ■

Remark 6.4 (Reality constraints and complex irreducibles). The theorem is complex-linear, but the molecule is real. Let C denote complex conjugation on the complexified displacement space. Since $CD = DC$ and $C\rho(g) = \rho(g)C$, conjugation carries the isotypic component of $V^{(i)}$ to that of its conjugate $\overline{V^{(i)}}$. In conjugate adapted bases the reduced blocks obey $F^{(\bar{i})} = \overline{F^{(i)}}$, so non-real conjugate irreducibles have the same spectrum and combine into real multiplets. A complex one-dimensional sector therefore need not give a nondegenerate physical mode.

In the brief real-type language: a self-conjugate complex irrep of real type has a real form; a non-self-conjugate irrep is of complex type and must travel with its conjugate; a self-conjugate irrep of quaternionic type has no real form and occurs with the corresponding even-multiplicity constraint when a real representation is complexified. We keep the complex character-table calculus, adding only these reality constraints.

The planar rotation representation of C_3 is the diagnostic example. On $\mathbb{R}^2$, let c act by $R(2\pi/3)$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. After complexification the representation is the sum of the conjugate one-dimensional characters $\chi_1 \oplus \chi_2$. A real matrix commuting with the rotation

has the form $aK + bJ$; if it is also symmetric, as a dynamical matrix is, then $b = 0$. Thus $D = aK$: the two conjugate complex singlets form a reality-enforced real doublet, not an accidental coincidence.

Three comments. First, the division of labour: *group theory fixes the block structure — the sizes m_i , the forced multiplet factors n_i , the symmetry-adapted bases — while the dynamics (spring constants, masses) only fills in the numerical entries of the small reduced matrices $F^{(i)}$* . A $3N$ -dimensional eigenvalue problem collapses into independent $m_i \times m_i$ problems. Second, a degeneracy larger than n_i can come from a repeated eigenvalue inside one $F^{(i)}$, from an accidental coincidence between unrelated blocks, or from the reality pairing of Remark 6.4; only the last is an additional physical constraint. Lecture 7 takes this theme into quantum mechanics. Third, the mechanism of the proof — an equivariant operator forced by Schur's lemma to be scalar across partner indices, leaving one reduced matrix element per pair of copies — is worth committing to memory: it is the Wigner–Eckart theorem in embryo.

The practical recipe, then:

- (1) identify the point group G of the equilibrium configuration;
- (2) compute $\chi_{\text{disp}}(g) = N_g \chi_{\text{vec}}(g)$ class by class;
- (3) subtract χ_{trans} and χ_{rot} ;
- (4) extract multiplicities $m_i = \langle \chi_i, \chi_{\text{vib}} \rangle_G$ from the character table;
- (5) diagonalise each reduced $F^{(i)}$: a reduced eigenvalue of multiplicity q gives complex degeneracy qn_i , then impose the conjugate-pair reality constraints;
- (6) construct explicit modes with the projectors of Lecture 5.

6.5 Worked example: water

Water is bent: $N = 3$, point group $C_{2v} = \{E, C_2, \sigma_v, \sigma'_v\}$ (Lecture 1). We place the molecule in the yz -plane with the C_2 axis along z ; σ_v denotes the molecular plane (yz) and σ'_v the perpendicular plane (xz). The group is abelian of order 4, so its four irreducible characters are one-dimensional sign characters; with Mulliken labels:

C_{2v}	E	C_2	$\sigma_v(yz)$	$\sigma'_v(xz)$	
A_1	1	1	1	1	z
A_2	1	1	-1	-1	
B_1	1	-1	1	-1	y
B_2	1	-1	-1	1	x

(the last column records how the coordinate functions transform, needed shortly; beware that interchanging the two planes interchanges the labels $B_1 \leftrightarrow B_2$ — conventions differ between books, so fix one and stay consistent).

Step 1: characters. The operation counts and traces:

	E	C_2	$\sigma_v(yz)$	$\sigma'_v(xz)$
N_g	3	1	3	1
χ_{vec}	3	-1	1	1
$\chi_{\text{disp}} = N_g \chi_{\text{vec}}$	9	-1	3	1
χ_{trans}	3	-1	1	1
$\chi_{\text{rot}} = \det \cdot \chi_{\text{vec}}$	3	-1	-1	-1
χ_{vib}	3	1	3	1

($N_{C_2} = 1$: only the oxygen sits on the axis; $N_{\sigma_v} = 3$: the whole molecule lies in its own plane; $N_{\sigma'_v} = 1$: only the oxygen lies in the perpendicular plane.)

Step 2: multiplicities. With all class sizes 1 and $|G| = 4$:

$$\begin{aligned} m_{A_1} &= \frac{1}{4}(3 + 1 + 3 + 1) = 2, & m_{A_2} &= \frac{1}{4}(3 + 1 - 3 - 1) = 0, \\ m_{B_1} &= \frac{1}{4}(3 - 1 + 3 - 1) = 1, & m_{B_2} &= \frac{1}{4}(3 - 1 - 3 + 1) = 0, \end{aligned}$$

so

$$\Gamma_{\text{vib}} = 2A_1 \oplus B_1,$$

three vibrational coordinates ($3N - 6 = 3 \checkmark$): two of full symmetry A_1 and one antisymmetric under both C_2 and σ'_v . Generically the 2×2 A_1 reduced block has two distinct eigenvalues, so all three frequencies are nondegenerate; an equal pair inside that block would be an accidental reduced-eigenvalue coincidence.

Step 3: explicit modes by projection. The vibrational physics is most transparent in internal coordinates: the two bond stretches s_1, s_2 (elongations of the O–H bonds) and the bend δ (the H–O–H angle increment) — three coordinates for three vibrations. The group permutes them simply: C_2 and σ'_v swap $s_1 \leftrightarrow s_2$, while E and σ_v fix each; δ is invariant under everything (hence pure A_1). The Lecture-5 projectors applied to the trial vector s_1 :

$$\begin{aligned} P_{A_1}s_1 &= \frac{1}{4}(s_1 + s_2 + s_1 + s_2) = \frac{1}{2}(s_1 + s_2), \\ P_{B_1}s_1 &= \frac{1}{4}(s_1 + (-1)s_2 + s_1 + (-1)s_2) = \frac{1}{2}(s_1 - s_2), \end{aligned}$$

with $P_{A_2}s_1 = P_{B_2}s_1 = 0$. The three symmetry-adapted vibrations of water (Figure 6.1):

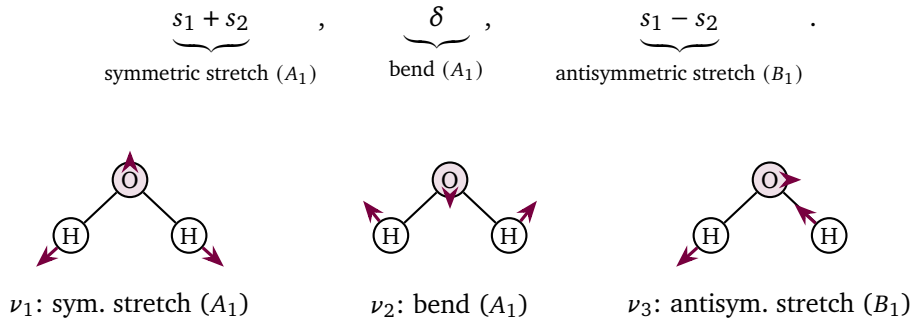


Figure 6.1: The three normal modes of water with their Mulliken labels. Small arrows on the oxygen keep the centre of mass fixed. The two A_1 coordinates mix; the patterns shown are the symmetry-adapted coordinates whose 2×2 reduced matrix determines the two A_1 frequencies.

Theorem 6.3 adds a final refinement that pure mode pictures gloss over: the two A_1 coordinates span a single isotypic component with $m_{A_1} = 2$, so the symmetric stretch and the bend are *not* individually normal modes — the actual eigenvectors are the two orthogonal combinations determined by the 2×2 reduced matrix $F^{(A_1)}$, whose entries depend on the force constants. The B_1 mode, alone in its symmetry type, is an exact normal mode by symmetry alone. Group theory has reduced a 9×9 problem to a 2×2 problem plus two trivia.

6.6 Worked example: ammonia

Ammonia is a triangular pyramid: $N = 4$, point group C_{3v} with the character table of Lecture 5,

C_{3v}	E	$2C_3$	$3\sigma_v$	
A_1	1	1	1	z
A_2	1	1	-1	
E	2	-1	0	(x, y)

(z along the threefold axis). The character bookkeeping, with class sizes 1, 2, 3:

	E	$2C_3$	$3\sigma_v$
N_g	4	1	2
χ_{vec}	3	0	1
χ_{disp}	12	0	2
χ_{trans}	3	0	1
χ_{rot}	3	0	-1
χ_{vib}	6	0	2

($N_{C_3} = 1$: nitrogen on the axis; $N_{\sigma_v} = 2$: nitrogen plus the one hydrogen contained in each vertical plane). Multiplicities, with $|G| = 6$:

$$\begin{aligned}
 m_{A_1} &= \frac{1}{6} [1 \cdot 1 \cdot 6 + 2 \cdot 1 \cdot 0 + 3 \cdot 1 \cdot 2] = 2, \\
 m_{A_2} &= \frac{1}{6} [6 + 0 - 6] = 0, \\
 m_E &= \frac{1}{6} [1 \cdot 2 \cdot 6 + 2 \cdot (-1) \cdot 0 + 3 \cdot 0 \cdot 2] = 2,
 \end{aligned}$$

so

$$\Gamma_{\text{vib}} = 2A_1 \oplus 2E :$$

six vibrational dimensions ($3N - 6 = 6$ ✓) and generically four distinct frequencies — two nondegenerate A_1 frequencies and two doubly degenerate E frequencies. This is exactly the observed infrared spectrum of ammonia: the symmetric stretch and the umbrella mode (A_1), and the doubly degenerate asymmetric stretch and asymmetric bend (E). The double degeneracies are not numerical accidents to be checked against force fields; they are theorems.

For explicit modes, internal coordinates again serve best: the three bond stretches s_1, s_2, s_3 and the three inter-bond angle increments $\delta_{12}, \delta_{23}, \delta_{31}$ — six coordinates for six vibrations. Under $C_{3v} \cong S_3$ each triple is simply permuted: the rotations cycle the indices, the reflection through H_1 swaps $2 \leftrightarrow 3$. But the permutation representation of S_3 on three objects is an

old friend: its character is the fixed-point count $(3, 0, 1)$, and by the one-line computation of Lecture 5 (or by inspection of the C_{3v} table above) it decomposes as

$$(3, 0, 1) = A_1 \oplus E.$$

Each triple therefore contributes one A_1 and one E , giving $2A_1 \oplus 2E$ in agreement with the Cartesian count — and the projectors are *verbatim* those computed for the S_3 permutation representation in Lecture 5, $P_{A_1} = J/3$ and $P_E = \mathbb{I} - J/3$ on each triple:

$$\begin{array}{ccc} \underbrace{s_1 + s_2 + s_3}_{\text{symmetric stretch } (A_1)} & , & \underbrace{\delta_{12} + \delta_{23} + \delta_{31}}_{\text{umbrella } (A_1)} \\ \underbrace{(2s_1 - s_2 - s_3, s_2 - s_3)}_{\text{asymmetric stretch, } E \text{ partners}} & , & \underbrace{(2\delta_{23} - \delta_{31} - \delta_{12}, \delta_{31} - \delta_{12})}_{\text{asymmetric bend, } E \text{ partners}} \end{array}$$

As with water, the two A_1 coordinates mix through a 2×2 reduced matrix, as do the two E pairs (partner by partner, with the *same* 2×2 matrix for both partners — that sameness is the content of Theorem 6.3(i) and the origin of the symmetry-enforced doublet factor). If the two eigenvalues of this reduced block happened to coincide, the total E degeneracy would be four. A 12×12 eigenvalue problem has become two 2×2 problems.

Remark 6.5 (Internal coordinates: a caveat). For water and ammonia the natural internal coordinates number exactly $3N - 6$ and the bookkeeping closes. For other molecules (rings, high-coordination centres) natural internal coordinates can be *redundant*; the character calculus detects this automatically — the internal-coordinate representation then properly contains Γ_{vib} , and the surplus must be identified and discarded. The Cartesian route of Sections 6.2–6.3 is never ambiguous.

6.7 Infrared and Raman activity: a preview

Which vibrations does a spectrometer actually see? Light couples to the molecular dipole moment, whose components transform like the coordinate functions (x, y, z) ; light scattering (Raman) couples to the polarisability, whose components transform like the quadratics $(x^2, y^2, z^2, xy, yz, zx)$. The full justification is the selection-rule machinery of Lecture 7, but the rule itself can be stated now:

A vibrational mode is infrared active iff its irreducible representation occurs among those of (x, y, z) , and Raman active iff it occurs among those of the quadratics.

For water, the table in Section 6.5 shows (x, y, z) spanning $B_2 \oplus B_1 \oplus A_1$; all three vibrations ($2A_1 \oplus B_1$) match and are infrared active — as observed, water absorbs at all three fundamentals (the reason it is a greenhouse gas). For ammonia, (x, y, z) spans $A_1 \oplus E$, so all four fundamentals $2A_1 \oplus 2E$ are infrared active — also as observed. The rule has teeth in molecules with more symmetry: in centrosymmetric molecules (point group containing the inversion i) the coordinates are odd under i while the quadratics are even, so *no mode is both infrared and Raman active* — the mutual exclusion rule, which Lecture 7 will derive in two lines from parity and which Sheet 6 explores.

Remark 6.6 (Computer algebra: numerics against theorems). The accompanying notebook closes the loop between group theory and dynamics. It models water and ammonia with

harmonic bond and angle springs, builds the Hessian by symbolic differentiation, mass-weights it, and calls Eigensystem. The numerical spectra display the predicted generic shapes — three distinct frequencies for water; four frequencies for ammonia of which two are double to machine precision — for generic spring constants. Tuned parameters can repeat a reduced eigenvalue or align unrelated blocks. Applying the projector matrices P_i of Lecture 5 (built from the displacement representation (6.2)) to the numerical eigenvectors confirms each one's symmetry label, and conjugating D into a symmetry-adapted basis displays the block structure of Theorem 6.3: the 9×9 and 12×12 dynamical matrices fall apart into the predicted small blocks, with the E blocks of ammonia appearing in identical pairs.

Summary.

- *Vibrations as a representation.* Nuclear displacements carry a (generally reducible) representation of the molecule's point group (eq. (6.2)).
- *Symmetry commutes.* The point group commutes with the dynamical matrix (eq. (6.3)), so each frequency eigenspace is a representation.
- *Modes by irreducible.* The complex irrep dimension is the symmetry-enforced multiplet factor; a reduced eigenvalue of multiplicity q has total complex degeneracy $q \dim V^{(i)}$, with conjugate sectors paired by reality (Theorem 6.3, Remark 6.4).
- *Mode counting.* The displacement character is read off the atoms a symmetry leaves in place (Lemma 6.1); removing the translations and rotations (Lemma 6.2) leaves the vibrational content.
- *Symmetry-adapted modes.* Character projectors construct the normal-mode coordinates, as worked for H_2O and NH_3 .

Tutorial. Exercise Sheet 6.

References

Bishop, *Group Theory and Chemistry*; Tinkham, *Group Theory and Quantum Mechanics* (molecular vibrations); Elliott–Dawber, *Symmetry in Physics* (vibrational analysis).

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: 10.1016/0375-9601(75)90798-7.
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: 10.1103/PhysRevLett.35.1053.
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.